L10.1: The directional of steepest ascent/descent

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online BSc Program on Data Science

and Programming. This video is about the direction of steepest ascent and descent. So, the ideas

in this video will be based on what we have studied about gradients and directional derivatives.

So, let us start with this question about how to trace water flowing down the hill.

So, maybe we will do this with an example. So, for an electrification project, someone named Mohan Bhargava is trying to understand how water flows down a particular hill, the

name may be somewhat familiar. So how does water flow downhill? So, let us ask this question,

how does water flow downhill?

So, if you think about how it flows downhill it always moves in the direction where the

altitude decreases most rapidly. So, if you have a rock face, if you have something like

this, it is going to flow down here because this is most rapid. So, just as an example

here is the, here is a cross section of the Deccan Plateau around Bangalore and Chennai,

and so you can see Bangalore here in the middle, you can see Chennai here at very close to

sea level on the right and there is the western ghats on the left, and a little bit of inclination

between Bangalore and then a steep downfall towards Chennai.

So, just as an example, if you have water, which is somewhere over here it is going to

flow down here. If you have water on this side, it will flow down on this side. And let us say if you have water somewhere here it will flow down here, if you have water

here it will flow down here, if you are here it flows towards Chennai and so on. So, I

think it is clear what we mean by it will move in the direction where the altitude decreases

most rapidly.

Now, then, of course, one question. So, what happens if you have water somewhere over here?

Somewhere exactly at the top. And there it is actually kind of stable. So, if you have

gone to mountain tops you might often find that on the top of the mountain it is often

like a plateau and there is, often pools of water over there. So, that does not happen.

When you have, when you are on the edges. So, which say something about the rate of

change, if you think of it carefully about what is happening to the altitude. This picture

is just to give a general idea of what will happen to water. Let us come back to the example

of our friend Mohan Bhargava.

So, what Mohan Bhargava does is, he models the hill similar to the previous picture using

a function h of x, where h is the altitude. And then he calculates the derivative h prime

x and computes at which points it is negative, positive and 0 and that is exactly what we

meant when we said that it moves in the direction in which the altitude decreases most rapidly.

So, the decrease in the altitude will be captured by the rate of change of the altitude. And

so, that means h prime x is going to tell you what is happening at the point x or to

the altitude. And so, if the derivative is negative that means the water flows to the

right from that point. So, if we say it is negative that means the the the the tangent

line is like this. And if the tangent line is like this, it flows down here, which is

to the right.

And similarly, if the derivative is positive then the tangent line is like this. So, it

flows down here, that is to the left, and if the derivative is 0, then that means the tangent line is like this, and that means the water will remain stationary. And that

is exactly what the phenomenon that I was talking about, right at the tip of the mountain

at the top. So, this is what Mohan Bhargava tries to do, he tries to model the altitude

by this function h of x draws a picture similar to what we had before, although, not for the

entire cross section of India, but for that specific hill, and then then based on that

he tries to understand how water is flowing down.

So, Mohan s friend Gita points out that a one-dimensional model is probably not going

to help because the hill is two dimensional, so it has, it is like this and so a one-dimensional

picture is not going to help us because there are more than right and left, in real life

there are on the ground. That is there are many directions, not just right and left.

So, she uses a two-dimensional function h of x comma y to model the altitude h. And

here is an example of such a thing. So, you can see something like a hill, and this is the graph of a function and suitable exponential. And so, what she comes up with is a function

of this type h of x, y the actual hills have more contours and she models it using that

function h of x, y. And based on that she tries to understand how the water is going

to flow down.

So, Mohan asked her, how she will find out the direction in which the water flows, because

now there is more than two directions. So, if you had only a function of one variable

h of x then computing the derivative told you in what direction the water is going to

move, but now you have several directions. And how do I know now based on this function

how the water is going to flow down?

So, Gita says the same principle that the water will flow down along the steepest slope,

which is when the altitude decreases fastest. So again, the direction in which the altitude

decreases fastest. And now, how do we compute this? So now, rather than saying that we should take h prime, we know that if we want to know the rate of change in a particular direction,

we have something called the directional derivative.

So, she says that, to compute it, we have to find the direction in which the function

h decreases fastest, or equivalently, the rate of decrease of h of rate of decrease

of h, is fastest. So, this is the same as finding the vector u in which the directional

derivative hu is largest in absolute value amongst those for which it has negative sign.

So, hu should have negative sign, because we want to decrease, we wanted to decrease

h has to decrease fastest, so hu should have negative sign, but amongst those with negative

sign, those directions with negative sign, if you take the absolute value, it is the

largest.

So, in terms of the real line, this will be the value at which hu is smallest in which

we include the negative side, so hu must be smallest, that is what we are saying. So that is u such that hu is less than or equal to hv for all v in R2, so that is the smallest

value and we also demand that hu is negative, because if it is smallest and positive then

that is not going to be of much help. So, this is what we want. So, Gita has now given

a different way of doing the same thing, and a more realistic way of doing the same thing.

So, now we come to the main point of this video, which is, how do we know that direction?

So, now we will generalize this. So, suppose we have a function of n variables and it is

defined on a domain D in Rn, containing some open ball around the point a tilde, so we

will give a general answer and then we will come back and answer it in this specific question

of what direction will the water flow in.

So, we want to know with this hypothesis in what direction is the directional derivative

minimized? So, suppose gradient f exists and is continuous on some open ball around the

point a tilde. So once this happens, we have this wonderful theorem that tells us that if you want to compute fu, then the way to do that is that fu is equal to gradient f

at that point a tilde dot with u. So, this is equal to the norm of gradient f at a tilde

times the norm at u times cosine of theta. Where what is theta?

So, where theta is the angle between the gradient at a tilde and u. We have two vectors, so

we have seen this in the linear algebra part that if you have the dot product then that

is norm of the first vector times norm of the second vector times cosine of theta, where

theta is the angle between those two vectors. Since it is a unit vector we know that norm

of u is 1, so this is the same as gradient of f at a tilde times cosine of theta. Now,

this function f is fixed, so the gradient is fixed and the gradient at a tilde in particular

is fixed and so its norm is fixed.

So, as you vary across the different directions what changes is theta. So, this is going to

change depending on what is theta. So, as theta changes fu is going to change. So as u changes theta changes, and based on that fu is going to change. So, where is this going

to be maximized and where is this going to be minimized?

So, now that depends on the values of cosine theta, and cosine theta we understand very,

very, very well. So, cosine theta is smallest, so cosine theta is minimized when theta is

equal to pi. So, in other words, so if you think of what that means, that means the angle

between u and the gradient vector is pi that means they are in opposite directions. If

gradient is pointing here then you must be pointing here. So that is u is pointing in

the direction opposite to gradient f at a tilde.

So, this minimum value therefore, the minimum value of fu is attained when u is equal to

minus gradient of f at a tilde by norm, because we always want a factor of u has to be of

norm 1. So, u is in the opposite direction means u is minus of the minus of gradient

it is in that direction, and then to scale it you have to divide by the norm to make it norm 1.

So, the minimum value of fu is attained when u is this and is equal to. So, now you can

substitute u to be this. So, if you do that or you can just say that cosine theta is minus

of this so it equals minus of norm of gradient of f at a tilde. So, the upshot of this is

that in general if you have the directional derivative, the direction in which it is minimized

when your function happens to have this property that gradient f exists and is continuous on

some ball around a tilde is in the opposite direction to gradient f at a tilde.

And so, the unit vector you have to that appears there is minus gradient f because that is

the opposite direction divided by its norm, because it has to be norm 1. And when you

substitute u to be that, which is the same as taking cosine theta is minus 1 in this

expression we will get minus of norm of gradient of f at a tilde.

So, now, this answers the question that was originally posed. So, if you had the water flow in the steepest direction downwards, it is going to happen at a gradient of h at

the point x comma y minus of that, in the minus of that direction. So, at each point

you should take minus gradient of f h at x, y and that will be the direction in which

the water moves.

So, one can ask the same question about when is it maximized and when is it, when does

it remain unchanged. So, based on our previous discussion, we are going to have the same

thing happening provided of course, that the hypothesis holds. So, assume the same hypothesis

as in the previous slide which means that the gradient exists and is continuous on a

open ball around a tilde. So, in that case, we will get there h, sorry fu is going to

be gradient of f at a tilde times norm of u, times cosine theta.

Now, if you want to maximize this, so it is maximized and cosine theta is maximized. Well,

first of all, I can get rid of norm u because it is always norm 1, times cosine theta it is maximized when theta is equal to 0, which means

u is in the same direction as gradient at a tilde. So,

in other words, u is equal to gradient of f at a tilde by its norm.

And this also justifies why we call it the gradient. So, the gradient tells you, the

word gradient tells you the slope at a point that is what the gradient means, in English

in colloquial language. And so, what we are getting is that the gradient actually tells

you that at a point what is the steepest slope. So, if you are standing at a particular point

on a mountainside the slopes could vary in different direction. So, the gradient tells

you the direction in which you have the steepest slope.

And we can also answer the second question, which is when does it remain unchanged? So,

it remains unchanged when the rate of change is 0, which means when fu is 0, so that is

when cosine is 0. And when is cosine 0? So cosine is 0 when theta is equal to pi by 2

that is theta is equal to pi by 2, which is the same as saying that u is orthogonal to or perpendicular because here we are using Euclidean norm and so on, so that is perpendicular

or at 90 degrees to the gradient vector.

So, we have answered these three things. When is it minimized in the opposite direction

to the gradient vector? When is it maximized when it is in the same direction as the gradient

vector? And when is the directional derivative constant when you move in a direction perpendicular

to the gradient vector.

Now, I want to warn you that in two dimensions, of course, saying perpendicular to the gradient

vector means that you go either, so it is a line, you go along this way or this way,

but we are in n dimensions. And in n dimension that means there is an equation that you have

to solve. And if you do your linear algebra, that means you have a subspace of dimension

n minus 1. And we will not talk a lot about that in this video, but this may come up later.

So, this is this the, these are the answers to these questions. So, we have completely answered the original question, which was in what direction does water flow? On the

other hand, if you want to ask, you are an adventure seeker and you want to go in the

direction where the slope is steepest, you want to do rock climbing and you want to climb

up fastest, so what direction should you go in, the direction of the gradient vector,

that is what we have understood from here.

And, on the other hand, if you have, if you do not want to change your altitude at all.

You want to keep going at the same altitude and just hope that you do not have to climb

down or climb up. And maybe if you go around then you will hit a plateau and you can go

you up, you do not want to do any climbing up or down. Then you go in the direction in

which the gradient is perpendicular, so you go in that direction. Fine.

So, just to summarize what we have said. So, assuming the hypothesis as in the previous

slide the property of steepest ascent is in terms of directional derivative that is saying that fu is positive and maximum and that takes place when u is equal to gradient of f divided

by its norm.

The property of steepest descent occurs in terms of directional derivative when fu is

negative and minimum and the direction is the negative of the gradient, and you divide

by its norm to make it norm 1. And the property of no change in the height in the altitude

or which means in terms of functions no change in the function. So that occurs in terms of

directional derivatives when fu is 0, and that is a direction orthogonal to gradient

of f. So, this summarizes what we have done so far.

Let us do a couple of examples to put this in perspective. So, suppose I have the function

f x, y is sin x y, let us compute the gradient. We have actually done this computation before,

so the gradient at x comma y is so at x you differentiate with respect to x, so that gives

me y times cosine of x, y, and then x times cosine x, y, that is with respect to y. So, let us choose a particular point and evaluate over there what happens.

So, suppose I choose the point, let us say pi comma 1, so at pi comma 1 what is the direction

of steepest descent on the graph of this function? So, let us first evaluate the gradient vector

at this point. So, if you take pi comma 1, so this is 1 times cosine of pi, comma pi

times cosine of pi.

Now cosine of pi is minus 1, so this tells us that this is minus 1 comma minus pi, so

the direction in which you should move is opposite to this. So, u is going to be minus

gradient f at pi comma 1 divided by its norm, which is 1 comma pi divided by root of 1 plus

pi squared. So, this is the direction in which you should move.

If you want to, for example, move in the direction where of steepest ascent, where the function

increases the fastest, the direction in which the function increases the fastest, then you

should move along minus 1 comma minus pi, the unit vector in that direction. And if you want to move along the direction where it does not change, so then you should move

perpendicular to this vector, which is the vector pi comma minus 1 or minus pi comma

1, so in either of those directions.

Let us similarly do this example. So, here we compute the gradient, so that is 2x comma

2y comma 2z. And so, suppose I want to now ask the same thing about let us say at 1 comma

1 comma 1 what is the direction in which the function increases fastest? So, the answer

is in the direction of the gradient vector.

So, the gradient at 1, 1, 1 is 2 comma 2 comma 2. And so, it should be in the direction,

so in the direction of u equals 2 comma 2 comma 2. And because we typically make this

a unit vector we should divide by its norm. So, 2 squared is 4 times 3 is 12, so this

is 1 over root 3, 1, 1, 1. So, instead if you wanted the direction of no change, so

in which direction does the function remain constant?

So, for this, we will have to move perpendicular to the vector 2, 2, 2. Now there is many choices

for the perpendicular direction. So, this will be any vector which is on the plane x

y z such that 2x plus 2y plus 2z is 0. So, I will encourage you to work out some vectors.

So, examples of such would be 1 comma minus 1 comma 0, of course, we should take the unit

vectors, so, I will divide by root 2 or you could have 1 by root 2 comma 1 comma 0 comma

minus 1.

And, in fact, this is a basis for those directions. So, any linear combination of these two would

be perpendicular to this direction, and that will be a direction in which the function

remains constant. So, I hope the idea here is clear. I want to emphasize that this this

video is extremely important because this is exactly what you are going to use when

you do something called gradient descent in machine learning.

So, in machine learning, you will have a phenomenon called gradient descent where you try to keep decreasing something or increasing something decreasing usually, and so, at a when you

are at a particular solution you try to minimize some quantity so as to get what we will call

a better solution. So, so please, yeah, please understand this video very well.

I want to end on this cautionary note, since we have been doing this example for a while,

now, I want to again emphasize this example to say that we need to know the continuity

of the gradient function. So, recall that for this function, the gradient of f at 0,

0 was 0 comma 0, but the directional derivatives at 0, 0 other than the x and the y axis meaning

the i and j these did not exist unless use e1 or e2 or minus e1 or minus e2, which will

be just negative of these.

So, this thing that we used earlier that fu is equal to gradient dot u, and which we use

in order to compute the direction in which it is maximum or minimum that here it is none

of those things are applicable. So, you need to know that your gradient is in fact continuous in a small neighborhood. And so, it may happen that the function is actually has edges.

So, it is jagged over there. And if that happens then you cannot use any of what we are doing

here. So, in such places, the gradient is not going to be, the partial derivatives will

not be continuous, and then we cannot apply this theory. So, it may still happen that

there is a direction. For example, if it is like this and inclination is more over here

you do not want to go in this direction, but we cannot say that from the gradient.

So, that condition is important. So, let us summarize quickly what we have done in this

video. So, we have seen this very important idea that, if you have a function f for which

the gradient functions continuous, then in order to compute the direction in which the

function increases most at a point that is the gradient at that point, the direction

in which the function decreases the most rapidly at that point, that is the negative of the gradient direction, and the direction in which the function stays constant that is orthogonal

or perpendicular to the direction of the gradient at that point. Thank you.